

Roll No.....

ODD MID SEMESTER EXAMINATION SEPTEMBER - 2025

Name of the Program: B. TECH

Semester: I

Name of the Course: Basic Electrical Engineering

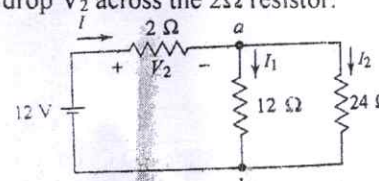
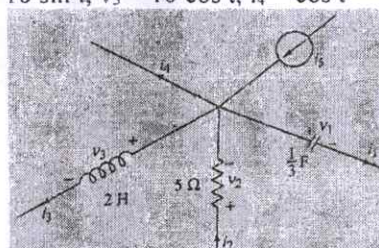
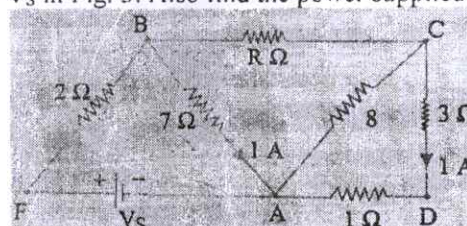
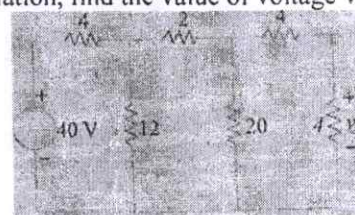
Course Code: TEE 101

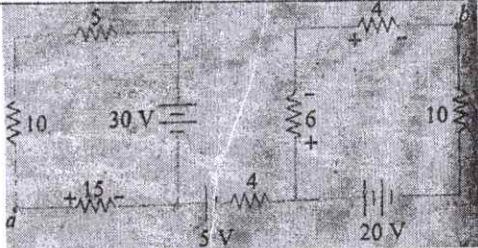
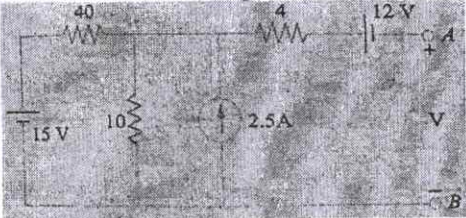
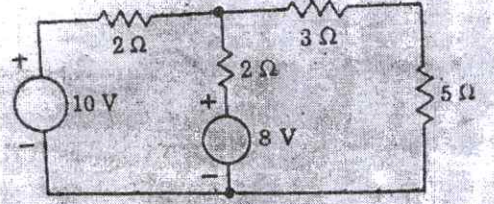
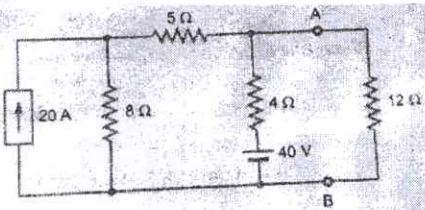
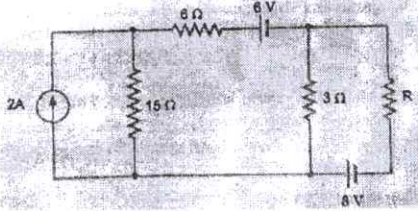
Time: 90 Minutes

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub questions.
- (ii) Each question carries 10 marks

Q1	(10 marks)	
a)	(i) Define properties of resistor (R), inductor (L), and capacitor (C) in an electrical circuit. (ii) In the circuit shown in Fig.1 calculate (a) currents I, I_1, and I_2; (b) the power consumed by each resistor; (c) the voltage drop V_2 across the 2Ω resistor.  Fig. 1	CO 1
OR		
b)	For the circuit shown in Fig. 2 determine the value of i_5 for the following values of voltages and current: $v_1 = 3 \sin t$, $v_2 = 10 \sin t$, $v_3 = 10 \cos t$, $i_4 = \cos t$  Fig. 2	
Q2	(10 marks)	
a)	Find the values of R and V_s in Fig. 3. Also find the power supplied by the source.  Fig. 3	CO 1
OR		
b)	Define following terms with suitable example: (i) Ideal and Practical voltage source (ii) Open circuit and short circuit condition (iii) Active and Passive elements (iv) Loop and Mesh (v) Voltage and Current Division Rule	
Q3	(10 marks)	
a)	(i) Explain different steps of Mesh analysis. (ii) By using source transformation, find the value of voltage v in the Fig. 4  Fig. 4	CO 2
OR		
b)	(i) Explain different steps of Node Voltage analysis. (ii) In the network shown in Fig. 5, calculate the voltage between points a and b i.e. V_{ab}.	

	 Fig. 5	
Q4	(10 marks)	
a)	(i) Explain Superposition theorem for a D.C. circuit. (ii) Using Superposition theorem, find the voltage V in Fig. 6  Fig. 6	CO 2
	OR	
b)	(i) Explain Thevenin's theorem for a D.C. circuit. (ii) Determine the current through 5Ω resistor in Fig. 7 by using Thevenin's theorem.  Fig. 7	CO 2
Q5	(10 marks)	
a)	(i) Explain Norton's theorem for D.C. circuit. (ii) By using Norton's theorem determine current flowing through 12Ω resistor for the network shown in Fig. 8.  Fig. 8	CO 2
	OR	
b)	(i) Explain Maximum Power Transfer theorem for D.C. circuit. (ii) Find the value of resistor 'R' to have maximum power transfer in the circuit as shown in Fig. 9. Also obtain the amount of maximum power.  Fig. 9	